

ROAD SAFETY INSIGHTS INDIA

A Power BI Dashboard Analysis of Road & Railway Accidents in India

Based on Official NCRB (Accidental Deaths & Suicides in India) Data

College Assignment Report — Power BI Data Analytics Project

Prepared using data.gov.in / NCRB source data

1. Introduction

Road and railway accidents remain one of India's most serious public safety concerns, causing hundreds of thousands of injuries and deaths every year. This project takes official Government of India accident data and turns it into an interactive Power BI dashboard so that patterns across states, union territories, and cities can be explored visually instead of read off static government tables.

The dashboard was built entirely on the NCRB's Accidental Deaths & Suicides in India (ADSI) report, Table 1A.2 — Incidence of Road and Railway Accidents by State/UT/City. This report documents the complete workflow that was followed, from raw data to the final interactive dashboard, and describes exactly what each part of the finished dashboard shows.

2. Objective of the Project

- Consolidate road accident, railway accident, and railway-crossing accident data into a single, clean, analysis-ready dataset.
- Present the national scale of accidents, injuries, and deaths through summary KPIs.
- Rank and compare states/UTs/cities by accident volume and severity.
- Compare road accidents against railway accidents visually, state by state.
- Calculate injury rate and death rate to measure the severity of accidents, not just their count.
- Build an interactive, filterable dashboard usable by transport planners, students, or the public.

3. Work Done — Project Workflow

The project followed the standard data analytics pipeline in Power BI: data collection, data cleaning, data modeling, DAX measure creation, and dashboard design.

3.1 Data Collection

The dataset used is the official NCRB ADSI Table 1A.2, covering 28 states, 8 Union Territories, and 54 major cities. It records, for each region, the number of Road Accidents, Railway Accidents, and Railway Crossing Accidents, along with the corresponding Cases, Persons Injured, and Persons Died for each category.

3.2 Data Cleaning (Power Query)

Before the data could be visualized, it was cleaned and prepared in Power Query:

- Data types were corrected — numeric columns (Cases, Injured, Died) set to Whole Number, and State/UT/City set to Text.
- The single raw table was separated into logical groups (States, UTs, and Cities) since city-level rows are already included inside their parent state's totals — this avoids double-counting when totals are calculated.
- Blank numeric cells were replaced with 0, since a blank in this dataset genuinely means zero recorded accidents of that type.
- Duplicate rows were checked and removed.
- State/UT names were standardized (for example, correcting the spelling of "Dadra and Nagar Haveli and Daman and Diu") for consistency.
- Unnecessary columns (like the raw serial number) were removed, and remaining columns were renamed for clarity (e.g., Road_Cases, Road_Injured, Road_Deaths).
- Calculated totals (Total Accidents, Total Injured, Total Deaths, Death Rate) were derived and validated against NCRB's own published All-India totals to confirm the cleaning did not distort the figures.

3.3 Data Modeling

A simple, purpose-built data model was used rather than an over-engineered one: a main Fact table holding state/UT-level accident figures, and a supporting City-level table linked to it for drill-down analysis. No date table was required since the source data is a single-year snapshot rather than a time series.

3.4 DAX Measures Created

Custom DAX measures were written to power the dashboard's KPIs and dynamic visuals, including:

- Total Accident Cases, Total Injured, Total Deaths — core aggregated totals.
- Death Rate % and Injury Rate % — deaths and injuries expressed as a percentage of total accident cases, to measure severity.
- State Rank by Cases / by Deaths — dynamic rankings that update as the user filters or selects states.
- Road Accident Share % and Railway Accident Share % — proportion of accidents by category.

3.5 Dashboard Design

The final dashboard was built as an interactive, multi-visual Power BI report with a consistent state/UT/city slicer panel, KPI summary cards, comparative bar charts, and a combined injury-and-death chart — described in detail in Section 4 below with the actual dashboard views.

4. What the Dashboard Shows — Screen-by-Screen

This section walks through the finished dashboard exactly as it appears, view by view.

4.1 National Summary KPIs

At the top of the report, four large KPI cards give an immediate national-level summary:

Metric	Value shown on dashboard
Total Accident Cases	541K
Total Injured	483K
Total Deaths	212K
Death Rate	39.21%
Injury Rate	89.32%

These cards are designed to be the first thing a viewer sees — giving the overall national scale of road and railway accidents before drilling into any state-level detail.

4.2 View 1 — Accident Cases by State/UT/City

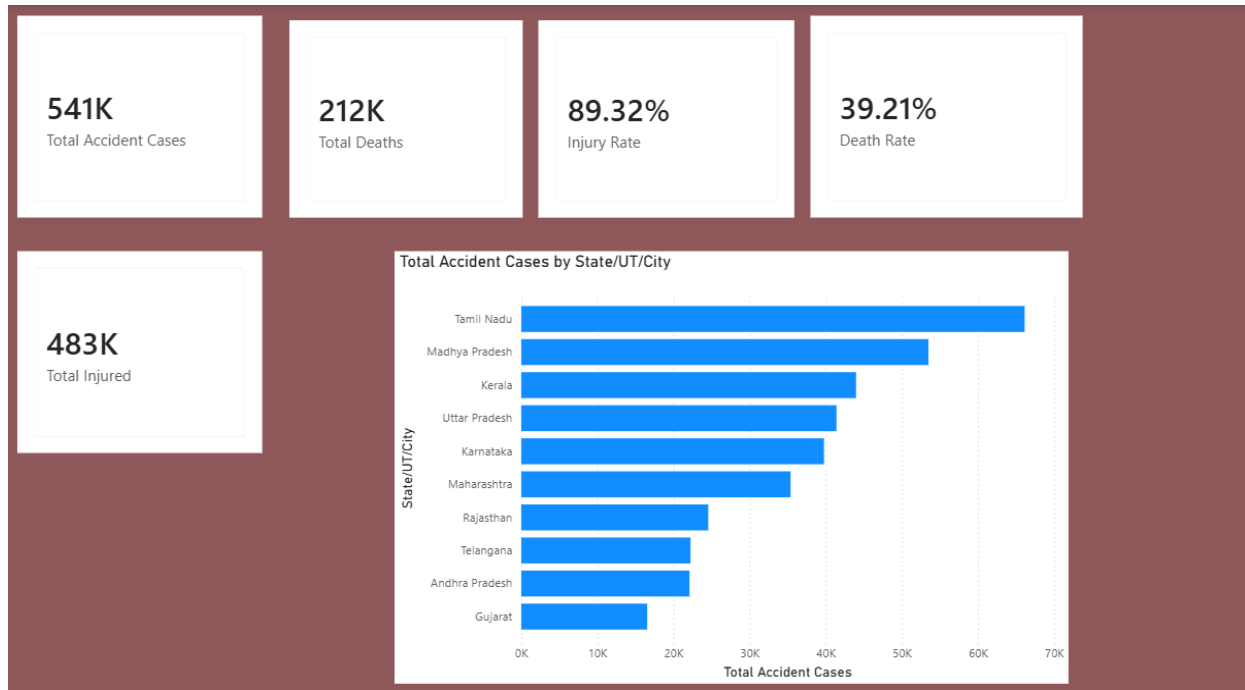


Figure 1: KPI cards followed by a horizontal bar chart of Total Accident Cases by State/UT/City, sorted highest to lowest.

This view combines the four core KPI cards (Total Accident Cases, Total Deaths, Injury Rate, Death Rate, Total Injured) with a horizontal bar chart ranking states by total accident cases. Tamil Nadu, Madhya Pradesh, and Kerala appear at the top, showing the highest accident volumes, while states like Gujarat and Andhra Pradesh appear lower on the top-10 list. This chart is the primary tool for identifying which states carry the largest accident burden.

4.3 View 2 — Railway vs. Road Accidents, and Injury/Death Comparison

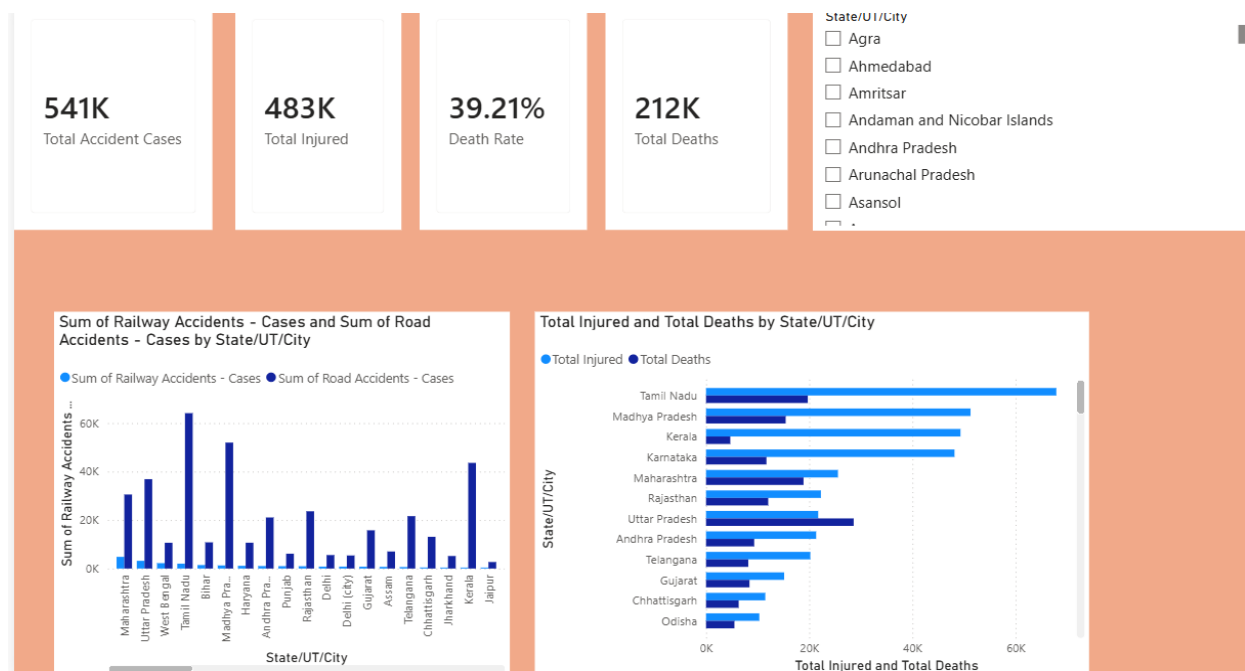


Figure 2: Combined view — KPI cards, a State/UT/City slicer, a Railway-vs-Road accident cases chart, and a Total Injured vs. Total Deaths comparison chart.

This is the main analytical page of the dashboard and contains three coordinated elements:

- A checkbox slicer panel (top-right) listing every State/UT/City — Agra, Ahmedabad, Amritsar, Andaman & Nicobar Islands, Andhra Pradesh, and so on — allowing the user to filter the entire page down to specific regions of interest.
- A dual bar chart titled "Sum of Road Accidents – Cases and Sum of Road Accidents – Cases by State/UT/City," which places Road Accident totals (light blue) and Railway Accident totals (navy) side by side for each state. This immediately shows that road accidents dominate the overall numbers, while railway accidents form a much smaller share — with Tamil Nadu and Madhya Pradesh again standing out as the states with the largest railway-accident bars relative to others.
- A horizontal Total Injured vs. Total Deaths chart by state, which shows that Tamil Nadu has the highest combined injured/death count, followed by Karnataka, Madhya Pradesh, and Maharashtra — useful for comparing whether a state's accidents tend to be more injury-heavy or more fatal.

4.4 View 3 — Reorganized Layout with Accident Analysis Tooltip

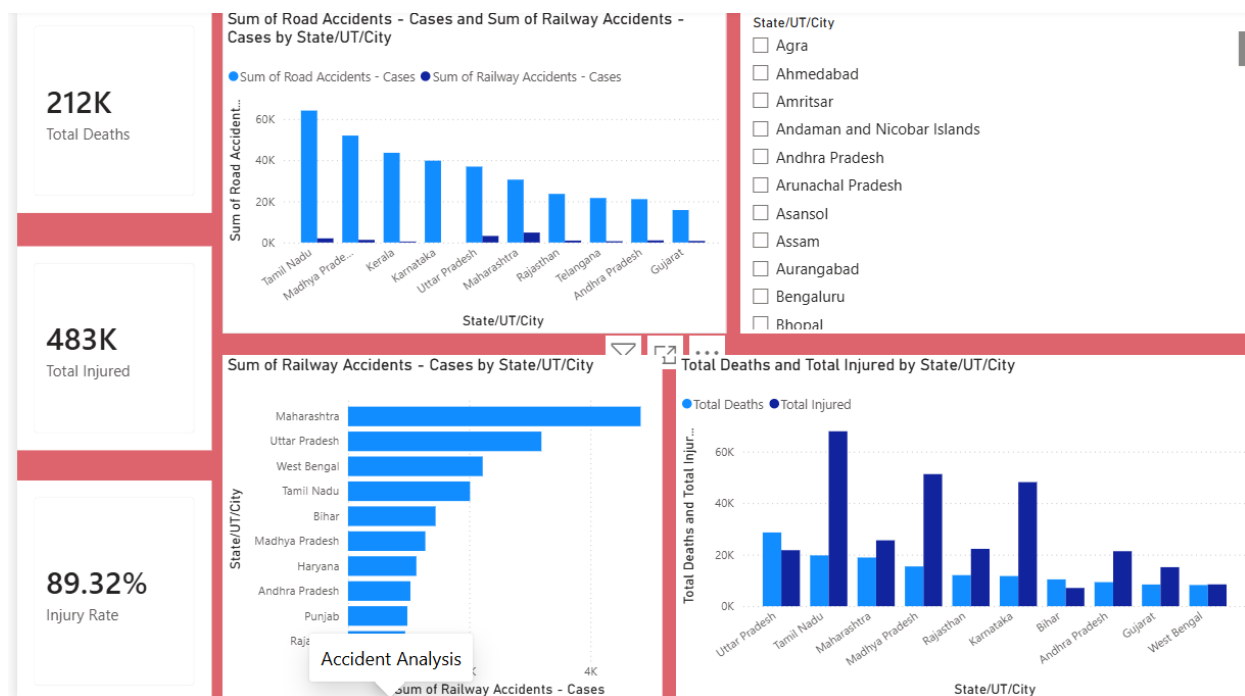


Figure 3: An alternate/expanded layout of the same visuals — KPI cards on the left, Road-vs-Railway accident chart, a state-wise Railway Accident ranking chart, and the Deaths-vs-Injured chart on the right, with an "Accident Analysis" tooltip label visible.

This view rearranges the same core visuals into a wider layout: the KPI cards (Total Deaths 212K, Total Injured 483K, Injury Rate 89.32%) sit on the left; the Road-vs-Railway accident comparison chart sits top-center; a dedicated "Sum of Railway Accidents – Cases by State/UT/City" ranking chart (Maharashtra, Uttar Pradesh, West Bengal, Tamil Nadu, Bihar leading) sits below it; and the Total Deaths vs. Total Injured comparison chart sits on the right, with the same full State/UT/City slicer available for filtering. The "Accident Analysis" label indicates this is the working title of this report page/tooltip within the Power BI file.

5. Key Observations from the Dashboard

- Road accidents make up the overwhelming majority of total accident cases nationally; railway accidents form a comparatively small share, though a few states (Tamil Nadu, Madhya Pradesh, Maharashtra) show a noticeably higher railway-accident count than the rest.
- Tamil Nadu consistently appears at or near the top across every chart — highest accident cases, highest combined injured/death totals — making it the single most significant state in this dataset.
- The national Injury Rate (89.32%) is far higher than the Death Rate (39.21%), meaning the large majority of people involved in recorded accidents are injured rather than killed — an expected but important severity signal.
- Maharashtra, Uttar Pradesh, and West Bengal stand out specifically on the railway-accident ranking chart, distinct from the states that lead on overall road-accident volume.

Note: These are the patterns visible directly in the current dashboard views. A deeper insights section (state-level severity ranking, top-5 national share, etc.) can be added once the double-counting check described in the earlier project guide is applied to the underlying data model.

6. Conclusion

This project successfully transformed a static official government table into an interactive Power BI dashboard capable of answering real analytical questions about road and railway accident patterns across India. The workflow — data cleaning in Power Query, a lean data model, custom DAX measures, and a multi-visual interactive report — demonstrates the complete Power BI analytics pipeline applied to authentic Government of India data, and provides a foundation for further work such as multi-year trend analysis or population-normalized comparisons.

7. Data Source

National Crime Records Bureau (NCRB), Ministry of Home Affairs, Government of India — "Accidental Deaths & Suicides in India (ADSI)" Report, Table 1A.2: Incidence of Road & Railway Accidents. Official source: ncrb.gov.in